

Q1)

ebay prod-vendor = (prod-id, model-num, vendor, delivery-storage, price)

FDs that hold on ebay prod vendor

prod-id $\rightarrow$ Price model-num

vendor storage $\rightarrow$ delivery-price

Decompose ebay prod-vendor:

prod-vendor-schema = (prod-id, price, model-num)

prod-vendor-info-schema = (prod-id, vendor, storage, delivery-price)

• Since prod-vendor-schema $\cap$ prod-vendor-info-schema = {prod-id}, it follows that our initial decomposition is a lossless decomposition as there is a common attribute

Decompose prod-vendor-info-schema:

prod-vendor-storage-schema = (vendor, storage, delivery-price)

prod-vendor-storage-info-schema = (prod-id, vendor, storage)

• Since prod-vendor-storage-schema $\cap$ prod-vendor-storage-info-schema = {vendor, storage}, it follows that our initial decomposition is a lossless decomposition as there is a common attribute.

Q2)

Let $R = (A, B, C, D)$

Let $F = \{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$ be a set of FDs

$C \rightarrow D$ $C \rightarrow A$

$C^+ = \{C, D, A\}$

C is not a candidate key as there is no B, therefore it is not a super key.
So, R is not in BCNF - there is a violation

$B \rightarrow C$ $C \rightarrow D$ $C \rightarrow A$

$B^+ = \{B, D, C, A\}$ B is a candidate key, therefore a superkey.

$C \rightarrow D$ and $C \rightarrow A$ is a non-trivial dependency, this cause a violation of BCNF. Therefore, replace R with two schemas

Decompose:

The non-trivial dependency

$R_1 = (C, D)$

C is a superkey of R_1 because $C \rightarrow D$

$R_2 = (A, B, C)$

B is a superkey of R_2 because $B \rightarrow C$

R_1 and R_2 are now in BCNF.